Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

JAVA ASSIGNMENT

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ROLL NO.:A018

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DATE OF SUBMISSION:06/05/24

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Kavya Gada

Experiment - 14

Aim: To implement basic Swing program with event handling.

⇒ Description of Experiment:-

① What is Java Swing?

Java Swing is a set of GUI (Graphical User Interface) components for Java programs. It allows developers to create rich, interactive user interfaces for their applications, including buttons, text fields, menus and more. Swing is a platform-independent meaning it can run on any operating system that supports Java.

Some features and concept of Java Swing are:-

ⓐ Platform Independent
ⓑ Lightweight Components
ⓒ MVC (Model View Controller) Architecture
ⓓ Rich Set of Components
ⓔ Layout Managers
ⓕ Event Handling
ⓖ Customization
ⓗ Internationalization

Overall, it provides robust framework for building desktop application with rich, interactive user interfaces.

② Swing Components:-

Common swing components used in building GUI applications are-

ⓐ JFrame: The main window of Swing application. It provides outer frame for containing other Swing components.

ⓑ JPanel: A lightweight container that can hold other components. It is often used to organize and group

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related components within a window

(c) JButton :- A clickable button that triggers an action when clicked by the user.

(d) JLabel :- A non-editable text component used to display a single line of text or an image.

(e) J-Text Area :- A multiline text area where users can enter and edit larger amounts of text.

(f) J-Check Box : A checkbox components that allows users to toggle text checked and unchecked states.

(g) JRadio Button : A radio button component that allows users to select one option from a group of mutually exclusive options

(h) JCombo Box :- A drop down combo box component that allows users to select an item from a list of options

(i) J Text Field :- A text input field where users can enter and edit text.

(j) JList :- A component that displays a list of item from which user can select one or more items.

(k) J Scroll Pane : A scrollable container that wraps around another component to provide scrolling capabilities when the content exceeds the visible area.

(l) Jslider : A slider component that allows users to select a value from continuous range by dragging a slider thumb.

(m) JProgress Bar : A progress bar component that visually indicates the progress of a task or operation.

(n) JFile Chooser : A component that allow user to choose files or directories from the file system.

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Above are commonly used components of Swing, there are many more available for building interfaces in Java Application.

(5) Event Handling in Swing with various listeners and their methods: Event Handling in Swing involves ~~Event Handl~~ registering event listeners to components and defining methods to handle specific events.

→ Some common swing listeners and their methods are: —

(a) ActionListener: Triggered when an component is activated,
- Method : actionPerformed (ActionEvent e);

(b) MouseListener: Triggered when mouse action occurs, such as clicks, presses, etc.
- Method : mouseClicked (MouseEvent e);
mousePressed (MouseEvent e);
mouseReleased (MouseEvent e); etc

(c) KeyListeners: Triggered when keyboard keys are pressed, released or typed.
- Method : keyPressed (KeyEvent e);
keyReleased (KeyEvent e); etc

(d) FocusListener: Triggered when a component becomes the focus or loses the focus.
- Method : focusGained (FocusEvent e);
focusLost (FocusEvent e);

(e) ItemListener: Triggered when the state of an item changes i.e. from unchecked to checked.
• Method : itemStateChanged (ItemEvent e);

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(f) ChangeListener : Triggered when state of component changes i.e. the value of Slider is adjusted.
• Method : StateChanged (ChangeEvent e);

(g) WindowListener : Triggered when window related events occur i.e. opening, closing etc.
• Method : windowOpened (WindowEvent e);
windowClosed (WindowEvent e); etc.

These Listeners allow Swing developers to create responsive and interactive user interfaces by handling various actions and components events.

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1) Write java program to create a registration form. Take login id and Password from the user and display it on the third Text Field which appears only on clicking Ok button and clear both the Text Fields on clicking RESET button:

Code :—

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

class Loginevent extends JFrame implements ActionListener {
    JFrame jf;
    JLabel l1, l2;
    JTextField tf1, ot;
    JPasswordField tf2;
    JButton b1, b2;

    Loginevent() {
        // creat component
        jf = new JFrame ("Login Test");
        tf1 = new JTextField (10);
        l1 = new JLabel ("Username");
        ot = new JTextField (20);
        l2 = new JLabel ("Password");
        tf2 = new JPasswordField (10);
        b1 = new JButton ("Ok");
        b2 = new JButton ("Reset");
        // set layout
```

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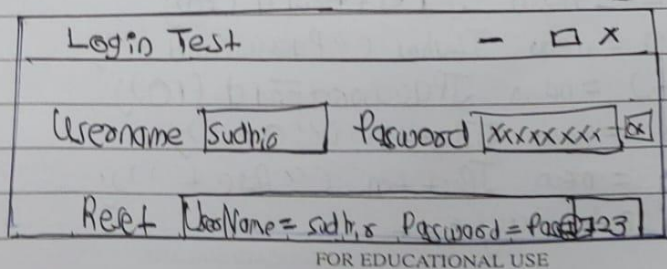
```
jf.setLayout(new FlowLayout());
// add component on frame /window
j.add(l1); jf.add(tf1); jf.add(l2);
jf.add(tf2); jf.add(b1); jf.add(b2); jf.add(ot);

//register component which are causing events
b1.addActionListener(this);
b2.addActionListener(this);
jf.setSize(500,100);
jf.setVisible(true);

public void actionPerformed(ActionEvent e) {
    if (e.getSource() == b1) {
        String s1 = tf1.getText();
        String s2 = String.valueOf(tf2.getPassword());
        ot.setText("User Name = " +s1+ " Password = "+s2)
    } else if (e.getSource() == b2)
    { tf1.setText("");
      tf2.setText("");
      ot.setText("");  }  }

public static void main(String args[]) {
    LoginEvent x = new LoginEvent();  }}
```



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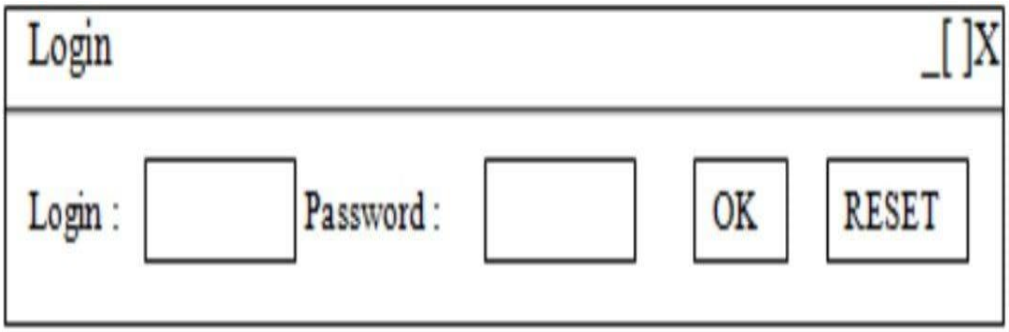
Experiment No: 14

Q1) Write a java program to create a registration form, Take Login and Password from user and display it on TextField which appears only on clicking OK button and clear both the TextField on clicking RESET button.



CODE:

```
import java.util.*;
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
```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```
class LoginEvent extends JFrame implements
ActionListener{
JFrame jf;
JLabel l1,l2;
JTextField tf1, ot;
JPasswordField tf2;
JButton b1,b2;
LoginEvent(){
//create componant

jf = new JFrame("Login Test");
tf1 = new JTextField(10);
l1 = new JLabel("Username");
ot = new JTextField(20);
l2 = new JLabel("Password");
tf2 = new JPasswordField(10);
b1 = new JButton("OK");
b2 = new JButton("Reset");
jf.setLayout(new FlowLayout());
jf.add(l1);
jf.add(tf1);
jf.add(l2);
jf.add(tf2);
```


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```
jf.add(b1);
jf.add(b2);
jf.add(ot);
b1.addActionListener(this);
b2.addActionListener(this);
jf.setSize(500,100);
jf.setVisible(true);
}


public void actionPerformed(ActionEvent e){
 if(e.getSource() == b1){
 String s1 = tf1.getText();
 String s2 = String.valueOf(tf2.getPassword());
 ot.setText("UserName = "+ s1+" Password = "+ s2);
 }
 else if(e.getSource() == b2){
 tf1.setText("");
 tf2.setText("");
 ot.setText("");
 }
 }
}
```


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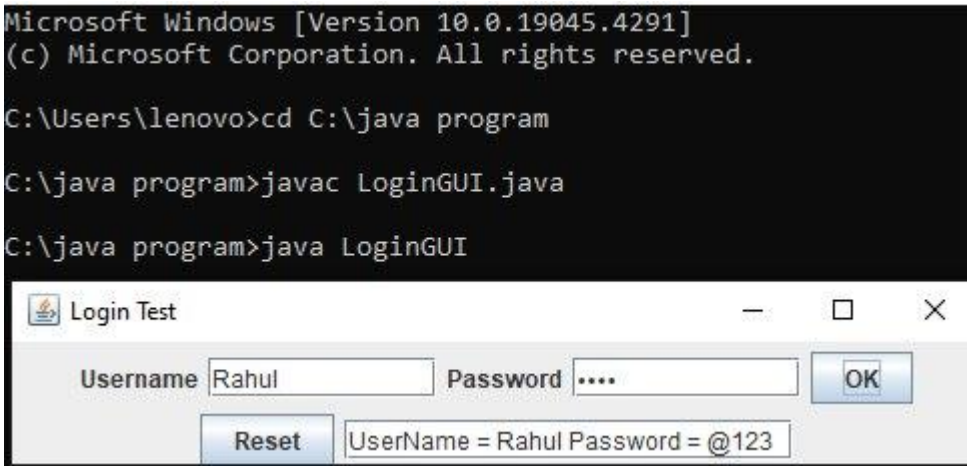
```
class LoginGUI{
 public static void main(String[] args){
 LoginEvent le = new LoginEvent();
}
}
```

OUTPUT:



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Q2) Write a program to create a basic calculator.

CODE:

```
import java.awt.event.*;
import javax.swing.*;
import java.awt.*;
class CalculatorNew extends JFrame implements
ActionListener{
 double a=0,b=0,result=0;
JTextField t;
JButton b0, b1, b2, b3, b4, b5, b6, b7, b8, b9, ba, bs, bd, bm,
be, beq, beq1;
```


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```
 int operator=0;
CalculatorNew(){
 b0 = new JButton("0");
b1 = new JButton("1");
b2 = new JButton("2");
b3 = new JButton("3");
b4 = new JButton("4");
b5 = new JButton("5");
b6 = new JButton("6");
b7 = new JButton("7");
b8 = new JButton("8");
b9 = new JButton("9");
// create operator buttons
beq1 = new JButton("=");
ba = new JButton("+");
bs = new JButton("-");
bd = new JButton("/");
bm = new JButton("*");
beq = new JButton("C");
be = new JButton(".");
t=new JTextField(20);
t.setSize(10,10);
JPanel p = new JPanel();
```


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```
bm.addActionListener(this);
bd.addActionListener(this);
bs.addActionListener(this);
ba.addActionListener(this);
b9.addActionListener(this);
b8.addActionListener(this);
b7.addActionListener(this);
b6.addActionListener(this);
b5.addActionListener(this);
b4.addActionListener(this);
b3.addActionListener(this);
b2.addActionListener(this);
b1.addActionListener(this);
b0.addActionListener(this);
be.addActionListener(this);
beq.addActionListener(this);
beq1.addActionListener(this);
 p.add(t);
p.add(ba);
p.add(b1);
p.add(b2);
p.add(b3);
p.add(bs);
```


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```
p.add(b4);
p.add(b5);
p.add(b6);
p.add(bm);
p.add(b7);
p.add(b8);
p.add(b9);
p.add(bd);
p.add(be);
p.add(b0);
p.add(beq);
p.add(beq1);
 setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
p.setBackground(Color.pink);
// add panel to frame
add(p);
 setSize(235, 235);
setVisible(true);
}
 public void actionPerformed(ActionEvent e){
if(e.getSource()==b1)
t.setText(t.getText().concat("1"));
if(e.getSource()==b2)
```


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```
t.setText(t.getText().concat("2"));
if(e.getSource()==b3)
t.setText(t.getText().concat("3"));
if(e.getSource()==b4)
t.setText(t.getText().concat("4"));
if(e.getSource()==b5)
t.setText(t.getText().concat("5"));
if(e.getSource()==b6)
t.setText(t.getText().concat("6"));
if(e.getSource()==b7)
t.setText(t.getText().concat("7"));
if(e.getSource()==b8)
t.setText(t.getText().concat("8"));
if(e.getSource()==b9)
t.setText(t.getText().concat("9"));
if(e.getSource()==b0)
t.setText(t.getText().concat("0"));
if(e.getSource()==be)
t.setText(t.getText().concat("."));
 if(e.getSource()==ba){
a=Double.parseDouble(t.getText());
operator=1;
t.setText("");
```


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```
}
if(e.getSource()==bs){
a=Double.parseDouble(t.getText());
operator=2;
t.setText("");
}
if(e.getSource()==bm){
a=Double.parseDouble(t.getText());
operator=3;
t.setText("");
}
if(e.getSource()==bd){
a=Double.parseDouble(t.getText());
operator=4;
t.setText("");}
if(e.getSource()==beq1){
b=Double.parseDouble(t.getText());
switch(operator){
case 1: result=a+b;
break;
case 2: result=a-b;
break;
case 3: result=a*b;
```


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```
break;
case 4: result=a/b;
break;
default: result=0;}
t.setText(""+result);}
if(e.getSource()==beq)
t.setText("");
 }

public static void main(String args[]){
 CalculatorNew cal=new CalculatorNew();}}
```

OUTPUT:

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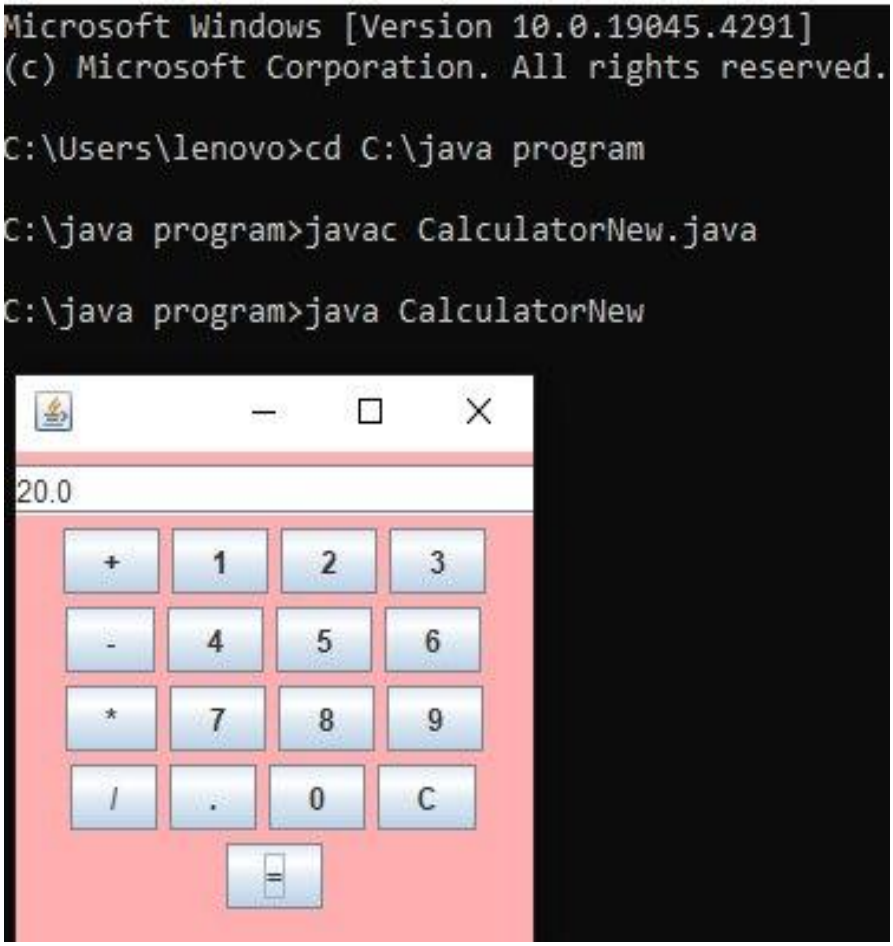

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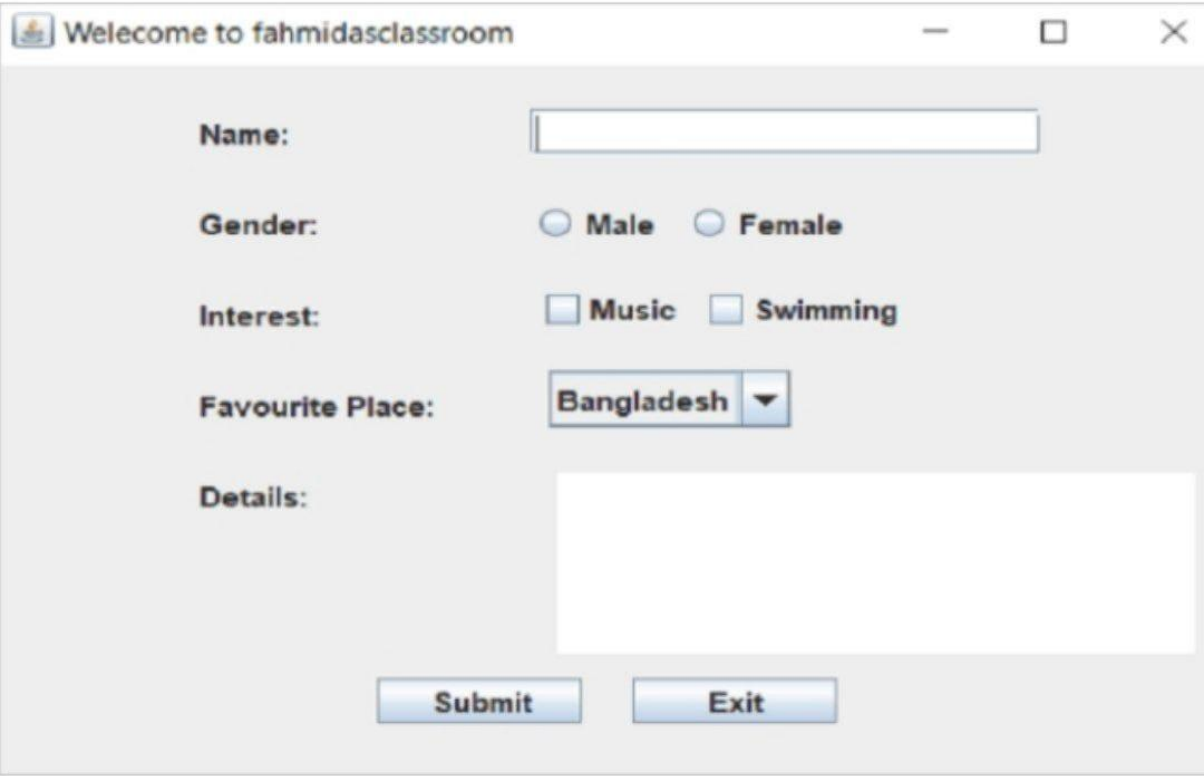
Q3)Display the selected fields in details after submit button is clicked.



CODE:

```
import java.awt.event.*;
import javax.swing.*;
import java.awt.*;
class Form extends JFrame implements ActionListener{
```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```
JLabel name,gen,interest,place,details;
JTextField tname;
JRadioButton male, female;
JComboBox cplace;
JButton submit,exit;
JTextArea tadd;
JCheckBox music,swim;
ButtonGroup gengp;

 public Form(){
 setTitle("Form");
setBounds(200, 80, 800, 800);
setDefaultCloseOperation(EXIT_ON_CLOSE);
setResizable(false);
 setLayout(null);
name = new JLabel("Name");
name.setFont(new Font("Arial", Font.PLAIN, 15));
name.setSize(100, 20);
name.setLocation(100, 100);
add(name);
tname = new JTextField();
tname.setFont(new Font("Arial", Font.PLAIN, 15));
tname.setSize(190, 20);
```


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```
tname.setLocation(200, 100);
add(tname);
gen = new JLabel("Gender");
gen.setFont(new Font("Arial", Font.PLAIN, 15));
gen.setSize(100, 20);
gen.setLocation(100, 150);
add(gen);
 male = new JRadioButton("Male");
 male.setFont(new Font("Arial", Font.PLAIN, 15));
 male.setSelected(true);
male.setSize(80, 20);
male.setLocation(200, 150);
add(male);
female = new JRadioButton("Female");
 female.setFont(new Font("Arial", Font.PLAIN, 15));
 female.setSize(80, 20);
female.setLocation(350, 150);
add(female);
gengp = new ButtonGroup();
 gengp.add(male);
 gengp.add(female);
interest = new JLabel("Interest");
interest.setFont(new Font("Arial", Font.PLAIN, 15));
```

```
interest.setSize(100, 20);
interest.setLocation(100, 200);
add(interest);
music = new JCheckBox("Music");
 music.setFont(new Font("Arial", Font.PLAIN, 15));
 music.setSize(100, 20);
 music.setLocation(200, 200);
 add(music);
swim = new JCheckBox("Swim");
 swim.setFont(new Font("Arial", Font.PLAIN, 15));
 swim.setSize(100, 20);
 swim.setLocation(350, 200);
 add(swim);
place = new JLabel("Fav Place");
place.setFont(new Font("Arial", Font.PLAIN, 15));
place.setSize(100, 20);
place.setLocation(100, 250);
add(place);
 cplace = new JComboBox();
 cplace.setFont(new Font("Arial", Font.PLAIN, 15));
 cplace.setSize(150, 20);
 cplace.setLocation(200, 250);
 add(cplace);
```


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```
cplace.addItem("India");
 cplace.addItem("USA");
 cplace.addItem("Bhutan");
 cplace.addItem("Srilanka");
 cplace.addItem("Maldives");

details = new JLabel("Details");
details.setFont(new Font("Arial", Font.PLAIN, 15));
details.setSize(100, 20);
details.setLocation(100, 300);
add(details);
 tadd = new JTextArea();
 tadd.setFont(new Font("Arial", Font.PLAIN, 15));
 tadd.setSize(200, 200);
 tadd.setLocation(200, 300);
 tadd.setLineWrap(true);
 tadd.setEditable(false);
 add(tadd);
 submit = new JButton("Submit");
submit.setFont(new Font("Arial", Font.PLAIN, 15));
submit.setSize(100, 20);
submit.setLocation(150, 550);
submit.addActionListener(this);
```


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```
add(submit);
exit = new JButton("Reset");
exit.setFont(new Font("Arial", Font.PLAIN, 15));
exit.setSize(100, 20);
exit.setLocation(270, 550);
exit.addActionListener(this);
add(exit);
setVisible(true);
}
public void actionPerformed(ActionEvent e){
if(e.getSource() == submit){
String data = "Name : " + tname.getText() + "\n" +
" Gender : " +(male.isSelected()==true?"Male":"Female") +
"\n" +
" Interests : "+ (music.isSelected()==true?"Music ":"")+
(swim.isSelected()==true?"Swimming":"") +"\n" +
" Favorite Place : " +
cplace.getItemAt(cplace.getSelectedIndex());
tadd.setVisible(true);
tadd.setText(data);
//tadd.setEditable(false);
}
else if(e.getSource() == exit){
```

```
tadd.setText("");
}
 }
}
class FormsGUI{
 public static void main(String[] args) throws Exception{
Form f = new Form();
}
}
```


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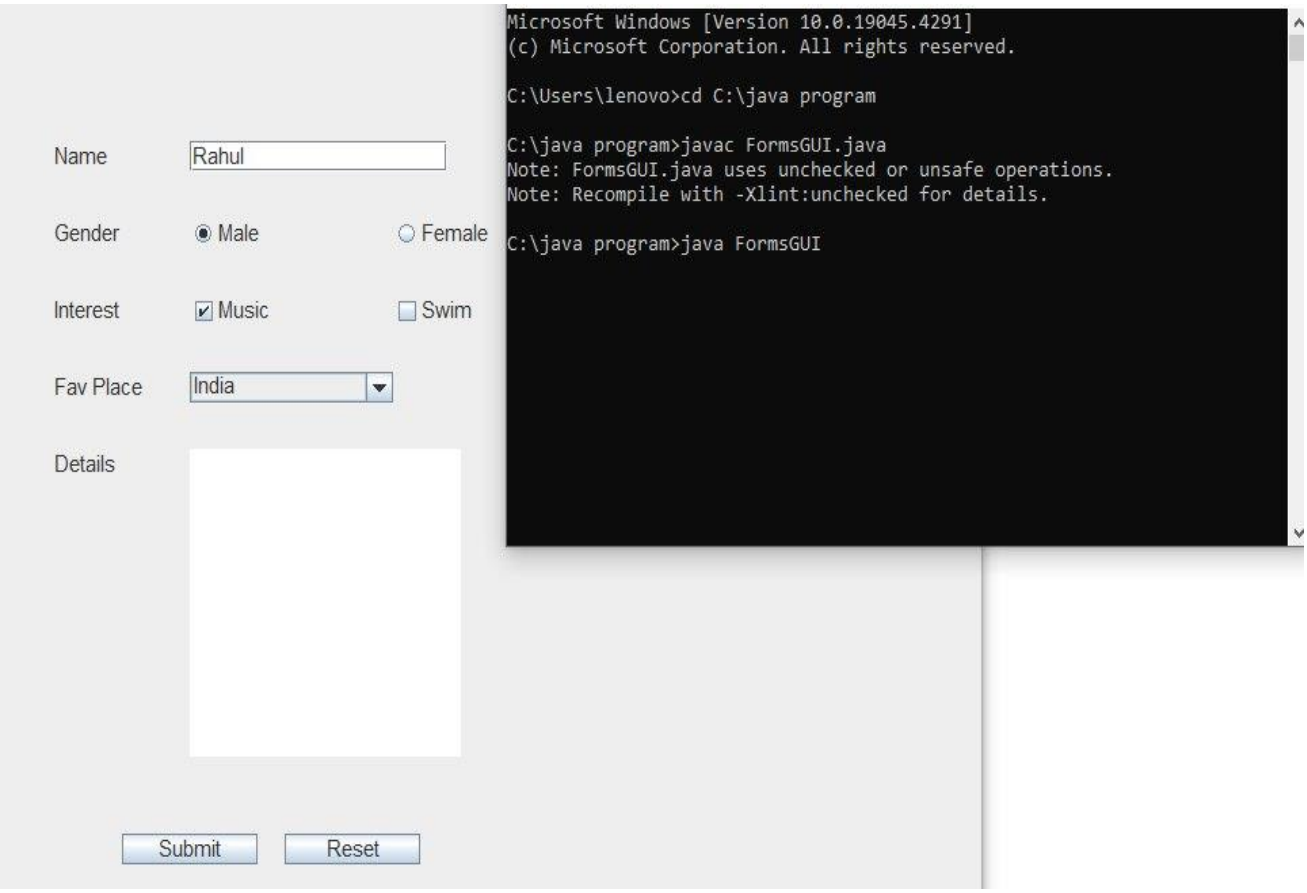
OUTPUT:



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CONCLUSION

We learnt about the implementation of java swing and awt.
We learnt what is gui and explored javax.swing
We also learnt about the Event handling and their various Listeners and their methods

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